



ECOCYCLE

Smart Waste, Clean Cities

A real, full-stack digital ecosystem connecting citizens, waste collectors and city authorities to make urban waste management transparent, circular and climate-smart — built for Kampala, Uganda.

UN SDG 11 · Sustainable Cities

UN SDG 12 · Responsible Consumption

UN SDG 13 · Climate Action

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1. Executive Summary

EcoCycle is a working software platform — not a mockup or a concept slide — that tackles one of Kampala's most dangerous and visible problems: solid waste. The city generates roughly **2,500 tonnes of waste every day**, yet only about half is ever collected. The rest burns in open pits, blocks drainage channels and feeds disasters like the **Kiteezi landfill collapse of 9 August 2024, which killed at least 35 people**.

EcoCycle attacks the problem at its root cause — a *broken information loop* between the people who produce waste, the people who collect it, and the authorities who must manage it. It unifies all three into a single mobile-first application with real accounts, real data, on-device artificial intelligence for waste sorting, gamified recycling incentives, digital payments, and a live city-impact dashboard.

The platform directly advances three United Nations Sustainable Development Goals (11, 12 and 13) and out-performs existing Ugandan and regional solutions on coverage, intelligence and citizen engagement.

~2,500t

waste generated daily in
Kampala

~48%

left uncollected each day

35+

killed in the 2024 Kiteezi
collapse

3-in-1

citizen · collector · city,
one app

2. The Problem — Kampala's Waste Crisis

Kampala Capital City Authority (KCCA) estimates the city produces about **2,500 tonnes of solid waste per day**, of which only around **1,200 tonnes are actually collected**. Official collection coverage has improved from roughly **35% in 2016 to 59% in 2023**, but that still leaves hundreds of tonnes per day in streets, wetlands and drainage channels.

For nearly three decades, almost all of that waste went to a single site: the **Kiteezi landfill**, opened in 1996 on about 36 acres, 12 km from the city centre. It reached its design capacity in **2012** but kept receiving 1,500–2,000 tonnes every day. On **9 August 2024, heavy rain triggered a collapse of the unstable garbage mountain** that buried homes and people — at least **35 confirmed dead and 28 missing**. The disaster forced the removal of KCCA's top leadership and exposed how fragile a one-landfill, low-visibility system really is.

Root cause: The crisis is not only physical, it is **informational**. Citizens have no easy channel to report illegal dumping or request a pickup; collectors work blind without digital job lists or proof of service; and city managers lack real-time data on where waste is accumulating. There is also almost **no incentive to separate or recycle at the source**, so recyclable plastic, organics and e-waste are dumped together and lost.

The gaps EcoCycle closes

- **No citizen reporting loop** — illegal dumps go unreported and unaddressed.
- **No source separation** — recyclables are mixed with general waste and buried.
- **Opaque collection** — no tracking, no proof, no accountability for pickups.
- **No behaviour-change incentive** — recycling is unrewarded, so it doesn't happen.
- **No real-time data** — authorities cannot target hotspots or measure progress.

3. Alignment with UN Sustainable Development Goals

EcoCycle is designed around three SDGs, mapping concrete product features to specific UN targets:

SDG	Specific target	How EcoCycle contributes
11 — Sustainable Cities & Communities	11.6 — Reduce the per-capita environmental impact of cities, with special attention to municipal waste management.	Citizen reporting + collector job-tracking + a city dashboard raise effective collection coverage and shrink the volume of uncollected, hazardous waste.
12 — Responsible Consumption & Production	12.5 — Substantially reduce waste generation through prevention, reduction, recycling and reuse.	On-device AI sorting + a points/rewards economy push citizens to separate plastic, organics, paper, glass and e-waste so they can be recycled instead of dumped.
13 — Climate Action	13.3 — Improve education and capacity on climate mitigation.	Diverting organics and recyclables from landfill reduces methane (a potent greenhouse gas); the in-app education hub and impact dashboard build climate literacy.

4. The Solution — EcoCycle

EcoCycle, tagged "*Smart Waste, Clean Cities*," is a single Progressive Web App (installable on phones and as an Android package) that serves **three roles in one unified ecosystem**:



Citizen

Report waste with a photo and GPS location, get instant AI sorting guidance, schedule and pay for pickups, earn points and rewards for recycling, and learn through the education hub.



Collector

Receive a live job queue, navigate to reports, upload photographic *proof of collection*, and share live truck location — turning informal collection into an accountable service.



City / Admin

Monitor reports and complaints on a live map, broadcast emergency alerts, and read an SDG impact dashboard that quantifies waste diverted and coverage achieved.

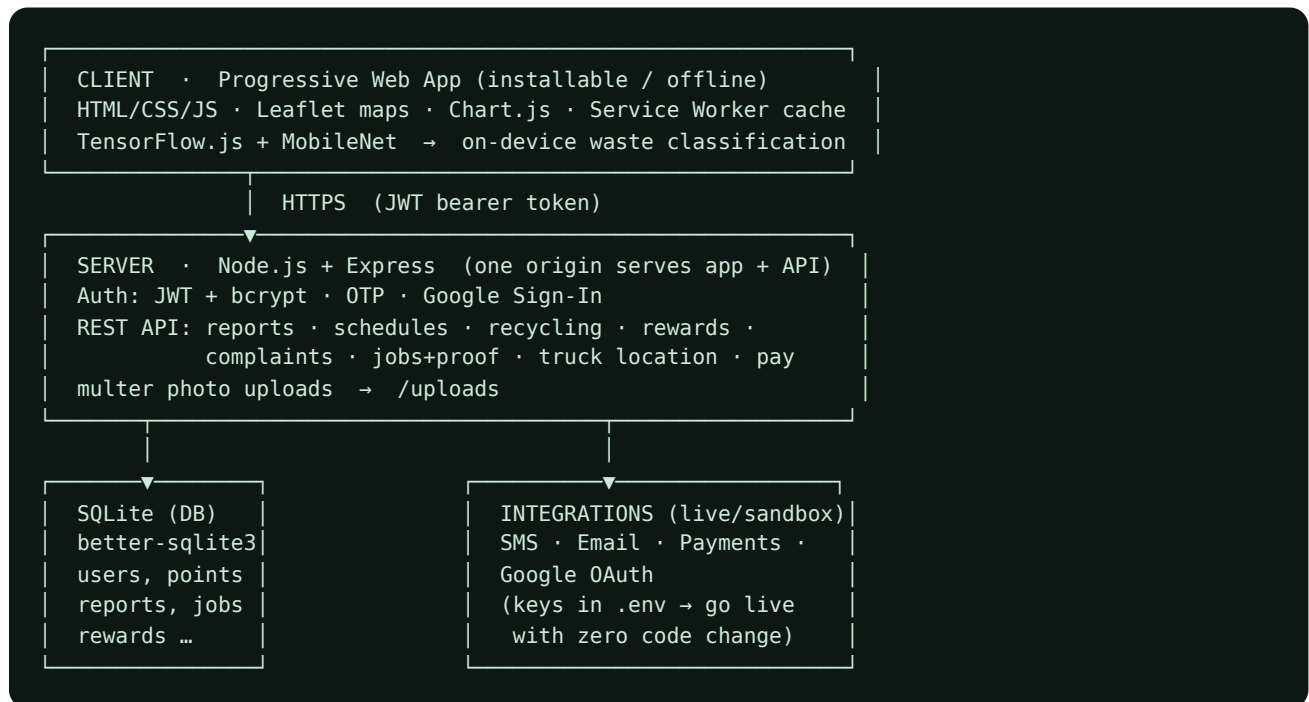


The loop

A citizen's report becomes a collector's job becomes a city data-point becomes a measurable SDG outcome — closing the information loop that the current system lacks.

5. System Architecture

EcoCycle is a **genuine full-stack application**: every feature is backed by a real server, a real database and real authentication. There are no fake screens.



Technology choices & why

Layer	Technology	Reason
Backend	Node.js + Express + SQLite (better-sqlite3)	Lightweight, zero-config database that runs anywhere — ideal for a low-resource deployment; one server process serves both the app and the API.
Auth & security	JWT, bcrypt, OTP, Google OAuth	Industry-standard hashed passwords and signed tokens; phone-based OTP suits a mobile-first Ugandan audience.
AI	TensorFlow.js + MobileNet (on-device)	Runs in the browser with no cloud cost and no privacy leak; works even on poor connectivity.
Frontend	Buildless PWA, Leaflet/OSM, Chart.js	Installable, offline-capable, and free of paid map keys — accessible on low-end Android phones.
Integrations	Africa's Talking / Twilio, SMTP, Flutterwave, Google	Local-first payment (Mobile Money via Flutterwave) and SMS; all toggle live ↔ sandbox via environment keys with no code change.

6. Key Features



Report & AI sort

Snap a photo of waste; on-device AI classifies it (plastic, organic, paper, glass, e-waste) and the report is geo-tagged for collectors.



Schedules & pickups

Citizens schedule collections and pay online (Mobile Money / card via Flutterwave), or use the sandbox flow for demos.



Points, levels & rewards

Recycling and reporting earn points and levels; points redeem for rewards — a behaviour-change engine for SDG 12.



Collector jobs + proof

Live job queue, navigation, and photographic proof-of-collection bring accountability to every pickup.



Live map & truck location

Leaflet/OpenStreetMap shows reports, hotspots and collection trucks in real time.



SDG impact dashboard

Charts quantify waste diverted, recycling rates and collection coverage for the city / admin role.



Emergency alerts

Authorities broadcast alerts (overflowing sites, hazards) directly to citizens.



Education hub

Guidance on sorting, composting and the circular economy — building climate capacity (SDG 13.3).



Works offline

A service worker caches the app shell and the AI model, so the core experience survives poor connectivity.



Circular economy

A waste-to-resource view frames collected material as input for compost, recycling and energy — not just garbage.

7. The Intelligence Layer — On-Device Waste Sorting

EcoCycle's standout technical feature is **artificial intelligence that runs entirely on the user's device**. Using TensorFlow.js with a MobileNet vision model (vendored locally, ~14 MB), the app classifies a photographed item and maps it to a waste category — for example plastic bottle → plastic , banana peel → organic , old phone → e-waste .

Why on-device matters here

- **No cloud cost** — classification is free, so it scales to millions of users without server bills.
- **Privacy-preserving** — photos never have to leave the phone to be sorted.
- **Works offline** — the model is cached, so sorting works with little or no signal — essential in Kampala's connectivity reality.

This turns every citizen's phone into a recycling assistant, directly enabling the source-separation that SDG 12.5 calls for.

8. Impact Model

EcoCycle creates impact through a reinforcing cycle rather than a single intervention:

1. **Visibility** — citizen reports surface uncollected and illegally-dumped waste that the current system never sees.
2. **Action** — those reports become trackable collector jobs with proof, raising effective collection coverage above today's ~59%.
3. **Diversion** — AI sorting plus rewards pull recyclables and organics out of the landfill stream, cutting both volume and methane.
4. **Measurement** — the city dashboard quantifies progress against SDG targets, enabling data-driven decisions.

Every recyclable diverted reduces pressure on overstretched sites like the one that collapsed at Kiteezi, and every tonne of organics kept out of landfill cuts greenhouse-gas emissions.

9. Competitive Analysis

EcoCycle was benchmarked against the leading waste solutions in Uganda and the region. Existing players solve *pieces* of the problem; EcoCycle is the only one that unifies citizens, collectors and the city with intelligence and incentives.

Capability	Yo-Waste (UG)	Homeklin (UG)	T-Bin (KE)	EcoCycle
On-demand pickup & payment	✓	✓	—	✓
Citizen civic reporting of dumps	—	—	—	✓
AI waste sorting (on-device)	—	—	partial (smart bins)	✓
Gamified recycling rewards	—	—	—	✓
Collector accountability (proof)	partial	—	—	✓
City SDG impact dashboard	—	—	—	✓
Works offline (PWA)	—	—	—	✓
Unified 3-role ecosystem	—	—	—	✓

Yo-Waste is Uganda's app-based pickup leader; Homeklin is a long-running collection/waste-to-energy company; T-Bin is a Kenyan smart-bin sensor system. A full benchmark is in the project's `COMPETITIVE.md`.

10. Security & Engineering

- **Password security:** bcrypt hashing — plaintext passwords are never stored.
- **Session security:** signed JWTs with a server-side secret; protected API routes verify the token.
- **Account verification:** OTP (one-time passcode) and Google Sign-In.
- **Safe uploads:** photo uploads handled by multer and served from an isolated path.
- **Resilience:** a service worker provides offline operation; the API and static app share one secure origin.
- **Configurable integrations:** SMS, email and payments switch from sandbox to live purely through environment variables — secrets stay out of the codebase.

11. Scalability & Future Work

- **Multi-city rollout** — the role model and data schema generalise to any municipality beyond Kampala.
- **IoT smart bins** — fill-level sensors can feed the same dashboard for predictive collection.
- **Waste-to-resource partnerships** — integrate composting, recycling and waste-to-energy off-takers to monetise diverted material.
- **Government & KCCA integration** — official adoption would turn citizen reports into a formal service channel.
- **Carbon credits** — quantified landfill diversion could be packaged as verifiable climate impact.

12. Conclusion

Kampala's waste crisis is a tragedy of broken information as much as of insufficient trucks. **EcoCycle** rebuilds that information loop in software that already works today — connecting every citizen, collector and city official around shared, intelligent, incentivised waste management. By advancing SDGs 11, 12 and 13 with a real, deployable, offline-capable platform, EcoCycle offers a credible, scalable path from disasters like Kiteezi toward genuinely smart, clean cities.

Appendix A — Running EcoCycle

Local: `cd server && node server.js` → open `http://localhost:4000/` (the Express server serves both the app and the API on one origin).

Demo accounts (password `ecocycle`):

Role	Login
Citizen	amina.n@gmail.com
Collector	j.okello@ecocycle.ug
City / Admin	s.mbabazi@ecocycle.ug

Appendix B — References

1. Economic Policy Research Centre — *Kampala garbage, an issue much wider than just Kiteezi*. eprcug.org
2. Wikipedia — *Kiteezi Landfill* (collapse of 9 August 2024; 35+ confirmed dead; established 1996; capacity reached 2012). en.wikipedia.org/wiki/Kiteezi_Landfill
3. Kaza, S. et al. — *What a Waste 2.0*, World Bank global waste database. datatopics.worldbank.org/what-a-waste
4. MDPI *Sustainability* 11(19):5523 — *Status of Waste Management in East African Cities ... Kampala City's Sustainability*.
5. KCCA collection-rate figures (35% in 2016 → 59% in 2023) and ~2,500 t/day generation, ~1,200 t/day collected, as reported in the above sources.